



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Electrical Engineering
Academic Year: 2025-26 (semester I)
In-Semester Examination / Evaluation II

Class: Second Year B.Tech.

Course Name: Electrical Machines-I
Fields

Course Code: PCC-EE201

Date: 14.10.2025

Time: 14:30 to 15:15

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Question Statement	Marks	CO	BT
Q.1	List the main parts of a D.C. machine with neat diagram.	4	CO1	BT1
Q.2	State and explain Fleming's right hand rule.	3	CO1	BT1, BT2
Q.3	A 4 pole, lap wound, d.c. generator has a useful flux of 0.07 Wb per pole. Calculate the generated e.m.f. when it is rotated at a speed of 900 r.p.m. with the help of prime mover. Armature consists of 440 number of conductors. Also calculate the generated e.m.f. if lap wound armature is replaced by wave wound armature.	4	CO4	BT3
Q.4	State the purpose of commutator and brush in d.c. machines?	2	CO1	BT1
Q.5	State the principle of operation of a d.c. motor and write the mathematical expression of the magnitude of the force experienced by the conductor.	2	CO1	BT1
Q.6	A shunt generator delivers load current of 450A at 230 V and resistances of shunt field and armature winding are 50 ohms and 0.03 ohms respectively. Calculate the generated e.m.f.	5	CO4	BT3

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Handle and understand static and rotating electrical machines of different types and ratings for installation as per requirement.
CO 2	Find out specific rating of AC and DC machines for different application.
CO 3	Develop understanding and applications of various types of electrical machines used in electrical vehicles.
CO 4	Analyse and evaluate different conditions where the static and rotating electrical machines can be used.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Electrical Engineering
Academic Year: 2025-26 (semester I)
In-Semester Examination / Evaluation II

Class: SY B.Tech.

Course Name: Electromagnetic Fields

Course Code: PCC-EE204

Date: 10.10.2025

Time: 14:30 to 15:15

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Question Statement	Marks	CO	BT
Q.1	Explain the method of images for solving electrostatic problems. Solve for the potential due to a point charge located near an infinite grounded conducting plane.	5	CO3	BT2, BT3
Q.2	Explain the significance of Laplace's equation in determining the potential distribution between the plates of a capacitor.	5	CO3	BT2
Q.3	Define magnetic flux density B . Using Maxwell's equations, explain the relationship between magnetic flux density and magnetic field intensity H in a linear, isotropic medium.	5	CO4	BT1, BT2
Q.4	Explain the concept of magnetic scalar potential and magnetic vector potential. Demonstrate the relation between magnetic vector potential A and magnetic flux density B .	5	CO4	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand the applications of vector algebra.
CO 2	Learn basic theory of electric and magnetic fields.
CO 3	Evaluate the Electrostatic boundary value conditions and problems.
CO 4	Analyse various aspects of magneto static fields.
CO 5	Understand magnetic forces materials and devices.
CO 6	Apply Maxwell's equations.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.

Department of Electrical Engineering
In-Semester Examination / Evaluation I
Academic Year: 2025-26 (semester I)

Class: SY

Course: Network Analysis/
Circuit Theory

Course Code: PCC-
EE203/PCCIN203

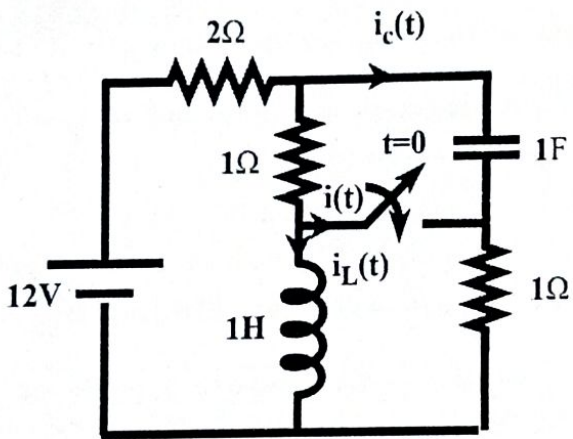
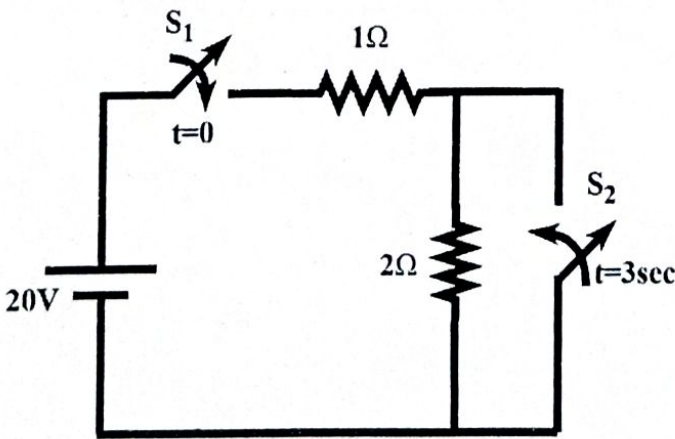
Date: 13.10.2025

Time: 14:30 to 15:15

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q.1	In the following Figure the switch is opened for a long time and is closed at $t=0$. Determine: $i(0^+)$, $dV_c(t)/dt$ at $t=0^+$ and $di_L(t)/dt$ at $t=0^+$ 	10	CO3	BT3
Q.2	For the circuit shown in the figure, the Switch S_1 is closed at $t=0$ and Switch S_2 is closed at $t=3$ sec. Determine $V_c(t)$ at $t=4$ sec. Plot $V_c(t)$ 	10	CO3	BT3



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Electrical Engineering
In-Semester Examination II
Academic Year: 2025-26 (Semester – I)

Class: Second Year B.Tech.	Subject: Analog Electronic Circuits	Subject Code: PCC-EE201
Date: 11-10-2025	Time: 14:30 to 15:15	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.

Q.N.		Marks	CO	BT
Q.1	Explain the structure of a MOSFET and its I-V characteristics.	5	CO1	BT1
Q.2	Describe how a MOSFET can be used as a switch.	5	CO2	BT1
Q.3	Briefly explain the small-signal model and biasing of a MOSFET when used as an amplifier.	5	CO3	BT1
Q.4	Define transconductance and describe its significance in a MOSFET amplifier.	5	CO4	BT1

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand the current voltage characteristics of semiconductor devices.
CO 2	Develop the ability to analyze and design analog electronic circuits viz. rectifier, amplifier, oscillator, regulator circuits using discrete components
CO 3	Analyze dc circuits and relate ac models of semiconductor devices with their physical Operation..
CO 4	Evaluate frequency response to understand behavior of electronic circuits.
CO 5	Understand the functioning of OP-AMP and design OP-AMP based circuits
CO 6	Take measurement of various analog circuits to compare experimental results in the laboratory with theoretical analysis.